

Ultrametric Quantum Error Correction

Testing Discrete Geometry in the Redundancy–Fidelity Tradeoff · CWI Summer School on QA & QEC · 24–28 August 2026 ·
Poster Session Wed 26 Aug

Core Idea

Standard QEC assumes noise accumulates over a continuous topology and fidelity scales smoothly with redundancy. We are testing an alternative: error propagation follows a discrete, hierarchical Bruhat–Tits tree geometry, where redundancy quantizes in units of $(p+1)$. **Signature:** a "staircase" of flat plateaus and sudden fidelity jumps instead of a smooth curve.
Hypothesis only — no result claimed.

Proposed Protocol (not yet run)

1. Encode one logical qubit in the Shor $[[9,1,3]]$ code.
2. Interact with a structured spectator-qubit environment.
3. Correct and measure final logical fidelity.
4. Group environment qubits into larger hierarchical fragments; track fidelity across fragment boundaries.

The Cross-Check Already Run — and It Mostly Failed (reported in full)

A 2-adic valuation witness ($v_p^{\max}$ of the Mahler spectrum) separates the Golay CSS $[[23,1,7]]$ code ($v_p^{\max} = 28$) from random codes (~ 4) — but does **NOT** generalize: Shor (2), Steane (2), Perfect (1), surface codes (2–6) all cluster at the random baseline. The gap is a Golay-symmetry artifact, not an optimality witness. **Open:** which witness can detect ultrametric structure?

Working Papers (Zenodo concept DOIs)

Ultrametric Code Spaces — 10.5281/zenodo.21824194
Archimedean Shadows (QEC–Darwinism tradeoff) — 10.5281/zenodo.21809888
Number-Theoretic Ultrametric Foundations (C7.3) — 10.5281/zenodo.21193003
ACRP-06: Extending $v_p^{\max}$ Code Classification — 10.5281/zenodo.21737221
Adelic Cross-Domain Program v5 — 10.5281/zenodo.21691414
"What Remains" (computing/QEC implications) — 10.5281/zenodo.21922812

Rowan Quni-Gudzinis · Independent Researcher · Adelic Physics Programme
papers.qnfo.org · zenodo.org/communities/qnfo